Riparian plant guilds: a community linked by space

Simon Frost

Table of contents

The model 1
Agreement with the original R code 2
What the wiring shows 3
Flow regimes 4
Reference 5

```
ENV["GKSwstype"] = "100"
using MatrixCommunityModels
using Plots, Random, Statistics
```

The model

Lytle et al. (2017) modelled five riparian plant guilds on the Yampa River, each with six stages: cottonwood (`C`, hydroriparian tree), tamarisk (`T`, xeroriparian shrub), willow (`W`, hydroriparian shrub), sagebrush (`S`, desert shrub) and meadow (`M`). Floods and droughts kill a stage-specific fraction of each guild. The dependency assumption is that seedlings can only establish on free space: the space occupied by all guilds after the year's mortality is subtracted from the total area `K`, and each guild's recruitment is proportional to what is left.

This differs from the fish model in two ways that the package has to express:

- recruitment does not depend on how many adults there are (one tree can seed all free space), so it is an additive `recruitment` term rather than a matrix entry;
- the space a guild occupies depends on the year, because it is counted after mortality, so its `occupancy` is a function of the state and the environment.

```
m = riparian_model()
m.community
```

```
OpenCommunity(C, T, W, S, M; pools = space, willow_adults)
```

Each year's environment is derived from one row of the 1916-1998 flow record at Maybell: whether there was a flood or drought, and the seed-release overlap and drawdown survival terms for cottonwood and tamarisk.

```
m.envs[1]
```

```
(year = 1916, flood = 1.0, drought = 0.0, nonflood = 0.0, cprop =
 ↪  0.9210526315789473, tprop = 0.26785714285714285, decline =
 ↪  0.6634536565168259, tdecline = 0.9063376969418638)
```

Agreement with the original R code

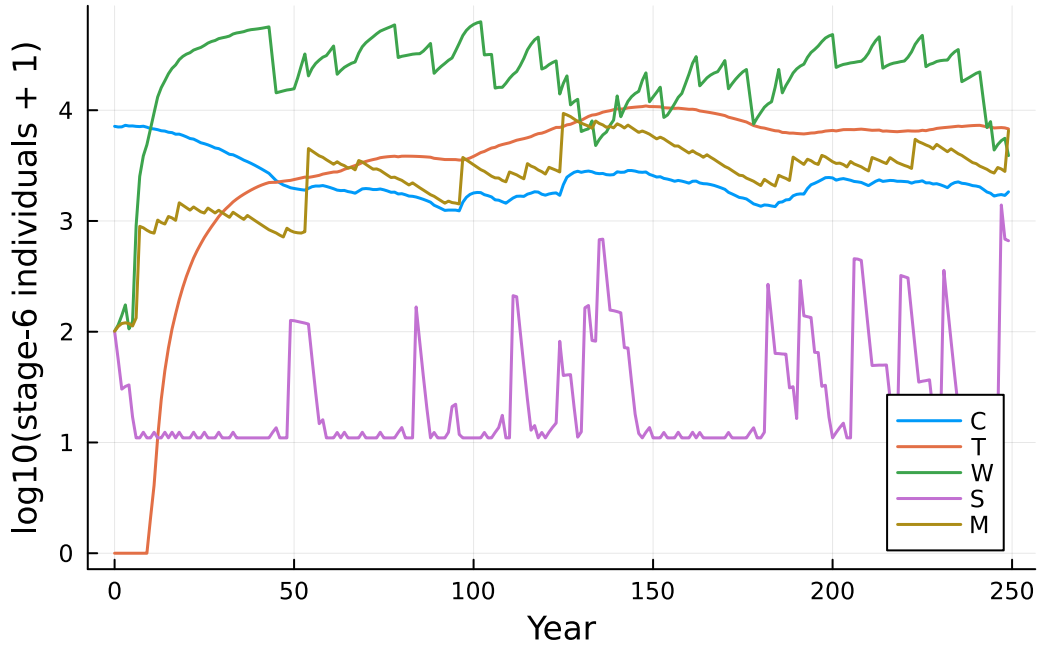
The published script draws a random year from the record at each step. The reference trajectory `rip_seed_traj.csv` was produced by that script with a fixed seed, and records which year was drawn, so the same sequence can be replayed here.

```
ref_file = joinpath(pkgdir(MatrixCommunityModels), "test", "reference",
 ↪  "rip_seed_traj.csv")
ref = permutedims(reduce(hcat, [parse.(Float64, split(l, ",")) for l in
 ↪  readlines(ref_file)[2:end]]))
drawn = Int.(ref[1:249, 2])
sol = project(m.community, m.n0, m.envs[drawn])
N = ref[:, 5:34]
(identical = count(sol.u .== N), of = length(N),
 max_relative_difference = maximum(abs.(sol.u .- N) ./ max.(abs.(N), 1.0)))
```

```
(identical = 7496, of = 7500, max_relative_difference =
 ↪  3.9221753509148656e-16)
```

Thirty states over 250 years agree to within a few parts in 10^16.

```
guilds = species_names(m.community)
mature = reduce(hcat, [sol[g][:, 6] for g in guilds])
plot(sol.t, log10.(mature .+ 1); label = permutedims(String.(guilds)),
 ↪  xlabel = "Year",
     ylabel = "log10(stage-6 individuals + 1)", lw = 1.5)
```



What the wiring shows

The original community has two pools, not one:

```
pool_names(m.community)
```

```
2-element Vector{Symbol}:
 :space
 :willow_adults
```

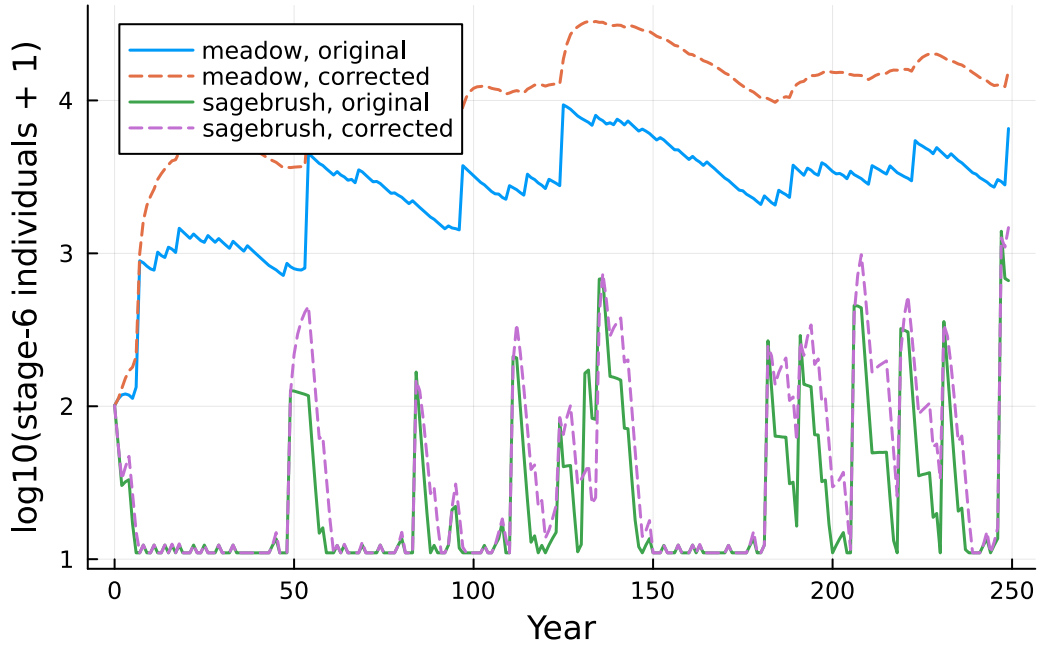
`space` is the dependency assumption. `willow_adults` is there because the published script assigns the willow guild's recruitment to the meadow guild, so meadow seedlings appear only when willow adults are present. In a script this is one variable name on one line; here a guild cannot see another guild except through a pool, so reproducing the behaviour required adding a wire, and the wire is visible.

The script also drops the stage-5 sagebrush and meadow individuals that do not mature, while still counting their space. `riparian_model(corrected = true)` removes both behaviours:

```
c = riparian_model(corrected = true)
pool_names(c.community)
```

```
1-element Vector{Symbol}:
 :space
```

```
corr = project(c.community, c.n0, c.envs[drawn])
plot(sol.t, log10.([sol[:M][:, 6] corr[:M][:, 6] sol[:S][:, 6] corr[:S][:,
 ↪  6]] .+ 1);
     label = ["meadow, original" "meadow, corrected" "sagebrush, original"
 ↪  "sagebrush, corrected"],
     xlabel = "Year", ylabel = "log10(stage-6 individuals + 1)", lw = 1.5,
 ↪  ls = [:solid :dash :solid :dash])
```



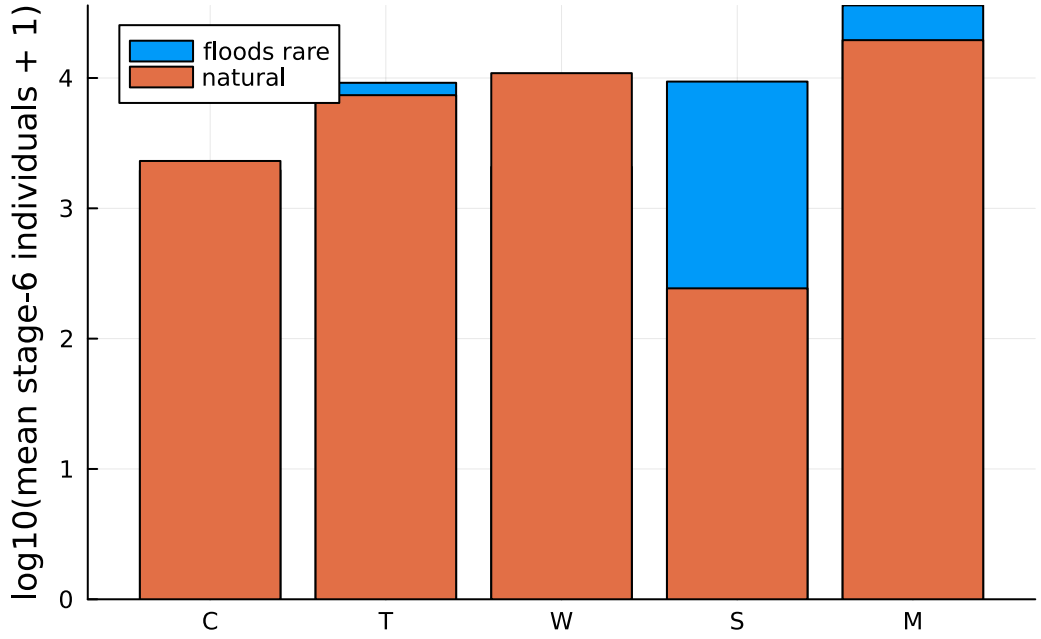
Both variants are kept: the original as a regression target, the corrected one as the model the paper describes.

Flow regimes

Projecting under a different flow regime only needs a different sequence of environments. Here the corrected model is run with years drawn at random from the whole record (floods in 57% of years) and with floods made rare (10% of years), comparing the mean number of stage-6 individuals of each guild after a 100-year burn-in. This is a single random sequence per regime, for illustration.

```
rng = Xoshiro(1)
floods = [e for e in c.envs if e.flood == 1]
others = [e for e in c.envs if e.flood == 0]
natural = [rand(rng, c.envs) for _ in 1:300]
fewer_floods = [rand(rng) < 0.1 ? rand(rng, floods) : rand(rng, others) for
 ↪  _ in 1:300]
```

```
function mean_mature(envs)
    s = project(c.community, c.n0, envs)
    [mean(s[g][101:end, 6]) for g in guilds]
end
a, b = mean_mature(natural), mean_mature(fewer_floods)
bar(repeat(String.(guilds); outer = 2), log10.([a; b] .+ 1);
    group = repeat(["natural", "floods rare"]; inner = 5), ylabel =
 ↪  "log10(mean stage-6 individuals + 1)")
```



Reference

Lytle, D. A., Merritt, D. M., Tonkin, J. D., Olden, J. D. & Reynolds, L. V. (2017). Linking river flow regimes to riparian plant guilds: a community-wide modeling approach. *Ecological Applications* 27, 1338-1350. Code and data: https://doi.org/10.6084/m9.figshare.4652608